

Yadong Liu

Curriculum Vitæ

School of Mathematical Sciences
Nanjing Normal University
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“Ein Mathematiker ist eine Maschine, die Kaffee in Theoreme umwandelt” - Alfréd Rényi

Personal Information

Gender Male
Birth November 27, 1996 (Yancheng, Jiangsu, P.R. China)
Nationality Chinese
Homepage <https://sites.google.com/view/yadongliu>

Appointment

2024.01 – Now **Lecturer, Nanjing Normal University**, Nanjing, China
2023.08 – 2023.12 **Wissenschaftlicher Mitarbeiter (Postdoc)**
University of Regensburg, Regensburg, Germany
Supported by startup funding of RTG 2339 *Interfaces, Complex Structures, and Singular Limits* of the German Research Foundation (DFG)
2020.10 – 2023.07 **Wissenschaftlicher Mitarbeiter (Research Assistant)**
University of Regensburg, Regensburg, Germany
In RTG 2339 *Interfaces, Complex Structures, and Singular Limits* of the German Research Foundation (DFG)

Education

2020.10 – 2023.07 **Dr. rer. nat. in Mathematics**
University of Regensburg, Regensburg, Germany
Dissertation: Analysis of fluid-structure interaction problems with growth
Grade: *magna cum laude*
Supervisors: Prof. Dr. Helmut Abels & PD Dr. Maria Neuss-Radu
Defense Committee: Bernd Ammann, Helmut Abels, Matthias Hieber, Harald Garcke, Bogdan Matioc.
2018.09 – 2020.06 **M.Sc. in Mathematics**
Nanjing University of Information Science and Technology, Nanjing, China
2015.09 – 2018.06 **B.Sc. in Mathematics and Applied Mathematics**
Nanjing University of Information Science and Technology, Nanjing, China
2014.09 – 2015.07 **Undergraduate study in Physics**
Nanjing University of Information Science and Technology, Nanjing, China

Research Interests

Analysis of nonlinear partial differential equations

- Fluid-structure interactions: bulk-surface, bulk-bulk, well-posedness, incompressible/compressible, singular limits
- Two-phase flows: diffuse interface models, viscoelastic fluids, well-posedness, long time dynamics, sharp interface limit

Publications & Preprints

◦ Preprints

15. H. Abels, M. Fei, **Y. Liu** and M. Moser, Higher Order Convergence for the Sharp Interface Limit of 3D Navier–Stokes/Allen–Cahn Systems, (2026) arXiv: 2606.30532.
14. M. Fei, X. Fei, **Y. Liu** and H. Wu, Global weak solutions to a diffuse-interface model for quasi-incompressible two-phase flows with unmatched densities and singular potential, (2026) arXiv: 2604.26660.

◦ Publications in Peer-Reviewed Journals

13. M. Fei, X. Fei, D. Han and **Y. Liu**, Weak solutions and incompressible limit of a quasi-incompressible Navier–Stokes/Cahn–Hilliard model for viscous two-phase flows, to appear at **SIAM J. Math. Anal.** (2025) arXiv: 2508.08090.
12. P. Knopf and **Y. Liu**, A Thermodynamically Consistent Free Boundary Model for Two-Phase Flows in an Evolving Domain with Bulk-Surface Interaction, to appear at **Interface Free Bound.** (2026) arXiv: 2507.01618.
11. **Y. Liu**, S. Mitra and Š. Nečasová, Weak solutions and singular limits for a compressible fluid-structure interaction problem with slip boundary conditions, **SIAM J. Math. Anal.** **58** (2026), no. 3, 2536–2577. DOI: 10.1137/25M1736827
10. M. Fei, X. Fei, D. Han and **Y. Liu**, Local-in-time existence of strong solutions to a quasi-incompressible Cahn–Hilliard–Navier–Stokes system, **Zeitschrift für Angewandte Mathematik und Mechanik** **106** (2026), no. 5, e70443, 1–24. DOI: 0.1002/zamm.70443
9. **Y. Liu** and D. Trautwein, On a diffuse interface model for incompressible viscoelastic two-phase flows, **J. Nonlinear. Sci.** **35** (2025), no. 12, 1–71. DOI: 10.1007/s00332-024-10103-w
8. H. Abels, **Y. Liu** and Š. Nečasová, Low mach number limit of a diffuse interface model for two-phase flows of compressible viscous fluids, **GAMM-Mitteilungen** (2024), e202470008, 1–15. DOI: 10.1002/gamm.202470008.
7. H. Abels and **Y. Liu**, Short time existence of a quasi-stationary fluid-structure interaction problem for plaque growth, **Adv. Nonlinear Anal.** **12** (2023), no. 1, 20230101. DOI: 10.1515/anona-2023-0101.
6. H. Abels and **Y. Liu**, On a fluid-structure interaction problem for plaque growth: cylindrical domain, **J. Differential Equations** **345** (2023), 334–400. DOI: 10.1016/j.jde.2022.11.046
5. H. Abels and **Y. Liu**, On a fluid-structure interaction problem for plaque growth, **Nonlinearity** **36** (2023), 537–583. DOI: 10.1088/1361-6544/aca5e1
4. Y. Zheng, W. Liu and **Y. Liu**, Equivalence between internal observability and exponential stabilization for suspension bridge problem, **Ric. Mat.** **71** (2022), no. 2, 711–721. DOI: 10.1007/s11587-021-00566-4
3. W. Liu, Y. Luan, **Y. Liu** and G. Li, Well-posedness and asymptotic stability to a laminated beam in thermoelasticity of type III, **Math. Methods Appl. Sci.** **43** (2020), no. 6, 3148–3166. DOI: 10.1002/mma.6108
2. **Y. Liu** and W. Liu, Blood flow analysis in tapered stenosed arteries with the influence of heat and mass transfer, **J. Appl. Math. Comput.** **63** (2020), no. 1-2, 523–541. DOI: 10.1007/s12190-020-01328-5

1. **Y. Liu** and W. Liu, Chaotic behavior analysis and control of a toxin producing phytoplankton and zooplankton system based on linear feedback, *Filomat* **32** (2018), no. 11, 3779–3789. DOI: 10.2298/fil1811779l

Academic Activities

◦ Research stays

- July–Aug. 2026 Tianyuan Mathematics Research Center, Kunming, China
Visiting Scholars Program: *Multiscale dynamics and singular limits of two-phase flows*
- Apr., May 2026 School of Mathematics and Statistics, Anhui Normal University, Wuhu, P.R. China
Host Professor: Prof. *Mingwen Fei*, Ph.D.
- June 2025 Fakultät Mathematik, Universität Regensburg, Regensburg, Germany
Host Professor: Prof. Dr. *Helmut Abels*
- Mar. 2025 School of Mathematical Sciences, Fudan University, Shanghai, P.R. China
Host Professor: Prof. *Hao Wu*, Ph.D.
- Jan., Nov., 2025 School of Mathematics and Statistics, Anhui Normal University, Wuhu, P.R. China
Host Professor: Prof. *Mingwen Fei*, Ph.D.
- Oct. 2024 School of Mathematics and Statistics, Anhui Normal University, Wuhu, P.R. China
Host Professor: Prof. *Mingwen Fei*, Ph.D.
- June 2024 Fakultät Mathematik, Universität Regensburg, Regensburg, Germany
Host Professor: Prof. Dr. *Helmut Abels*
- Nov. 2023 Institute of Mathematics, Czech Academy of Sciences, Prague, Czech Republic
Host Professor: RNDr. *Šárka Nečasová*, DSc
- Oct. 2023 Institute for Analysis, Karlsruhe Institute of Technology, Karlsruhe, Germany
Host Professor: JProf. Dr. *Xian Liao*
- Aug. 2023 School of Mathematical Sciences, Fudan University, China
Host Professor: Prof. *Hao Wu*, Ph.D.
- Nov. 2022 Institute of Mathematics, Czech Academy of Sciences, Prague, Czech Republic
Host Professor: RNDr. *Šárka Nečasová*, DSc
- Apr.–June 2022 Institute of Mathematics, Czech Academy of Sciences, Prague, Czech Republic
Host Professor: RNDr. *Šárka Nečasová*, DSc

◦ Organisation of meetings

- Sept. 2026 The 17th International Conference on *Free Boundary Problems*, Berlin, Germany
Co-organized the minisymposium: Fluid-Structure Interactions
- Oct. 2022 Annual Meeting of RTG 2339 *IntComSin*, Weltenburg
Co-organized and contributed talk
- Apr. 2022 Workshop on *Analysis of Fluid and Elastic Bodies Interactions*, Regensburg
Co-organized and contributed talk

◦ Invited talks

- Sept. 2026 Minisymposia: *Phase Transitions and Pattern Formation* and *Phase Field Methods in Real-World Applications* at the 17th International Conference on *Free Boundary Problems*, Berlin, Germany
- July 2026 Special Sessions *Phase-field models and their singular limits* and *Mathematical Insights into Phase-Field Models* at *The 15th AIMS conference*, Athens, Greece

- May 2026 Workshop on *PDEs and Stochastic PDEs in Material Science and Statistical Mechanics*, CUHKSZ, Shenzhen, China
- Jan. 2026 International Conference on *Variational Methods in Fluid and Solid Mechanics*, Tsinghua Sanya International Mathematics Forum, Sanya, China
- Nov. 2025 Workshop on *Nonlinear Partial Differential Equations*, Soochow University, Suzhou, China
- June. 2025 *Mathematics with Applications 2025*, Funchal, Madeira, Portugal
- Mar. 2025 *Mathematics Integrated Seminar*, School of Mathematical Sciences, Fudan University, China
- Oct. 2024 *Youth Seminar on Differential Equations*, Anhui Normal University, Wuhu, China
- July 2024 Workshop on *Partial Differential Equations and Its application*, Southwestern University of Finance and Economics, Chengdu, China
- Apr. 2024 *Youth Seminar on Differential Equations*, Anhui Normal University, Wuhu, China
- Nov. 2023 Seminar on *Partial Differential Equations*, Institute of Mathematics, Czech Academy of Sciences, Prague, Czech Republic
- Oct. 2023 Seminar of the Work Group *Nonlinear Partial Differential Equations*, Karlsruhe Institute of Technology, Karlsruhe
- Aug. 2023 *ICIAM 2023*, 10th International Congress on Industrial and Applied Mathematics, Waseda University, Tokyo, Japan
Minisymposium *Modeling and Computation for Interface Dynamics in Fluids and Solids*
- Aug. 2023 *Mathematics Integrated Seminar*, School of Mathematical Sciences, Fudan University, China
- Dec. 2022 SEU Yau Center *Geometry and PDE Weekend Seminar*, Nanjing, China (online)
- Apr. 2022 Working Seminar on Partial Differential Equations, Institute of Mathematics, Czech Academy of Sciences, Prague, Czech Republic
- **Participants and talks**
- Sep. 2026 School on *Fluids, Structures and their Interactions*, CIME, Cetraro, Italy
Poster
- Nov. 2024 Conference on *Mathematics of fluids in motion: Recent results and trends*, CIRM, Marseille Luminy, France
Small talk
- Oct. 2024 *The Seventh National Doctoral Forum on Partial Differential Equations*, Shenzhen University, Shenzhen, China
Contributed talk
- June 2024 *Equadiff 2024*, Karlstad, Sweden
Contributed talk
- May 2024 *The 5th International Symposium on Biological Mathematics and Medical Applications*, Nanjing, China
Contributed talk
- May 2024 *11th Youth Forum of Partial Differential Equations*, Wuhu, China
- Nov. 2023 International Workshop *Vector- and Tensor-Valued Surface PDEs*, Dresden
Contributed talk
- Oct. 2023 Oberwolfach Seminar: *Recent Topics on the Navier-Stokes Equations*, Oberwolfach

- Small talk
- Sept. 2023 11th GAMM Workshop on *Analysis of PDEs*, Eichstätt
Contributed talk
- Sept. 2023 Workshop on *Energetic Methods for Multi-Component Reactive Mixtures: Modelling, Stability, and Asymptotic Analysis*, WIAS Berlin
Contributed talk
- June 2023 Shocking Developments: *New Directions in Compressible and Incompressible Flows: A Conference in Honor of Alexis Vasseur's 50th Birthday*, Max Planck Institute for Mathematics in the Sciences, Leipzig
- May 2023 The GAMM Annual Meeting 2023, Dresden
Contributed talk in Session *Applied Analysis*
- Mar. 2023 Workshop on *Analysis of PDEs*, Karlsruhe
- Aug. 2022 Conference on *Mathematical Fluid Mechanics*, Prague, Czech Republic
Contributed talk
- Aug. 2022 The GAMM Annual Meeting 2022, Aachen
Contributed talk in Session *Applied Analysis*
- Aug. 2022 Hausdorff School on *Geometric Analysis and Nonlinear PDEs*, Bonn
Poster
- June 2022 Lake Como School of Advanced Studies *Mathematical Models for Bio-Medical Sciences*, Como, Italy
Contributed talk
- May 2022 International Symposium on *Biological Mathematics and Medical Applications*, NUIST, Nanjing, China (online)
Contributed talk at Graduate Forum
- May 2022 Conference *Nonlinear PDEs in Fluid Dynamics*, CIRM, Marseille Luminy, France
- Apr. 2022 Research school *Mathematical Advances in Geophysical Flows*, CIRM, Marseille Luminy, France
Small talk
- Mar. 2022 SIAM Conference on *Analysis of Partial Differential Equations*, Berlin (online)
- Mar. 2022 Workshop on *Mathematical Methods for Complex Phenomena*, Regensburg
Contributed talk
- Mar. 2022 Workshop on *PDEs for Surfaces and Interfaces*, Regensburg
- Oct. 2021 9th GAMM Workshop on *Analysis of PDEs*, Freiburg
- Oct. 2021 Annual Meeting of RTG 2339 *IntComSin*, Weltenburg (talk)
- Sept. 2021 International Conference *Free Boundary Problems*, Berlin (online)
- Aug. 2021 Summer School *Fluids under Control*, Prague (online)
- Aug. 2021 Germany-Japan Workshop on *Problems Related to Free Boundaries and Moving Contact Lines*, Tokyo (online)
- Oct. 2020 TES Student Compact Course *Variational Methods for Fluids and Solids*, Berlin (online)

Teaching

◦ Instructor

- Fall 26/27 Calculus I-1 (for NNU-Macquarie joint institute)

- Spring 26 Calculus I-2 (for NNU-Macquarie joint institute)
- Fall 25/26 Ordinary Differential Equations (for math students)
- Calculus I-1 (for NNU-Macquarie joint institute)
- Spring 25 Advanced Mathematics II-2 (for engineering students, only two weeks)
 - **Teaching Assistant**
- Fall 24/25 Mathematical Analysis III (for math students)
- Spring 24 Mathematical Analysis II (for math students)

Honors & Awards

- Nov. 2021 **Excellent academic degree master's thesis** of Jiangsu Province, Jiangsu Academic Degree Committee
- Oct. 2019 **National Scholarships for Graduate Students**, Ministry of Education of the People's Republic of China
- Oct. 2019 **First Prize**, Postgraduate Scholarship, Nanjing University of Information Science and Technology
- Oct. 2019 **National Second Prize**, 16th National Graduate Mathematical Modeling Contest
- June 2018 **Outstanding Graduates**, Nanjing University of Information Science and Technology
- Apr. 2017 **Meritorious Winner**, Mathematical Contest In Modeling/Interdisciplinary Contest In Modeling
- Oct. 2016 **National First Prize**, Contemporary Undergraduate Mathematical Contest in Modeling
- Nov. 2015 **First Prize**, Merit Scholarship, Nanjing University of Information Science and Technology

Languages & Skills

- Languages Chinese Mandarin (Native)
- Yancheng dialect (Native)
- English (Fluent)
- German (Elementary)
- Programing MATLAB, Mathematica, SPSS

References & External Links

References

- Prof. Dr. **Helmut Abels**
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- RNDr. **Šárka Nečasová**, DSc.
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Czech Academy of Science
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115 67 Praha 1, Czech Republic
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- Prof. Dr. **Harald Garcke**
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Universität Regensburg
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- PD Dr. **Maria Neuss-Radu**
Department Mathematik
FAU Erlangen-Nürnberg
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- Prof. **Wenjun Liu**
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Nanjing University of Information Science and Technology
Ningliu Road 219
Nanjing 210044, China
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○ External Links

- School of Mathematical Sciences, Nanjing Normal University
- Research Training Group 2339 (RTG 2339) *Interfaces, Complex Structures, and Singular Limits*
- GAMM Activity Group *Analysis of Partial Differential Equations*
- Fakultät für Mathematik, Universität Regensburg